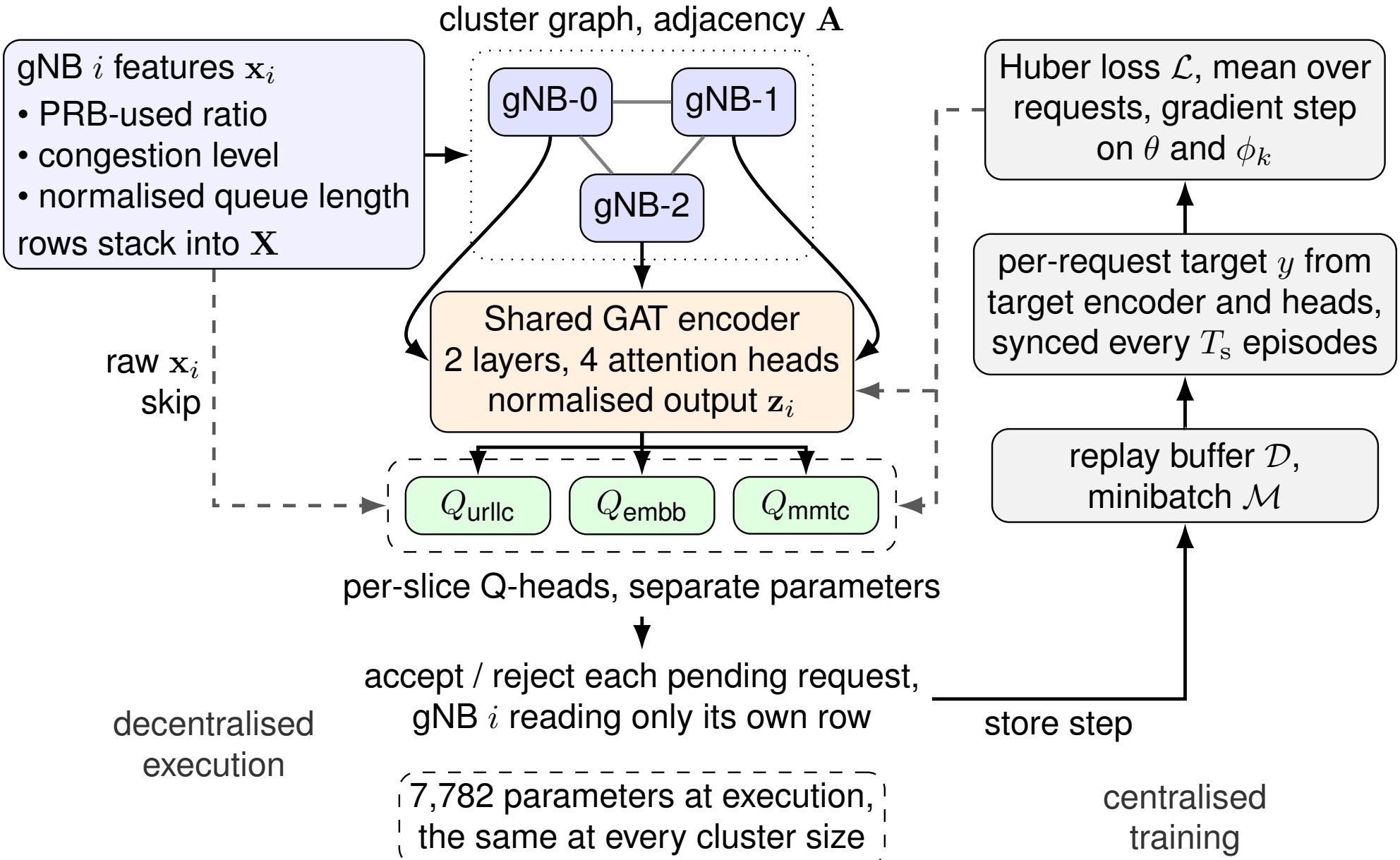

cluster graph, adjacency $\mathbf{A}$
gNB i features $\mathbf{x}_i$
• PRB-used ratio
• congestion level
• normalised queue length
rows stack into $\mathbf{X}$
gNB-0
gNB-1
gNB-2
Shared GAT encoder
2 layers, 4 attention heads
normalised output $\mathbf{z}_i$
raw $\mathbf{x}_i$
skip
Q_{urllc}
Q_{embb}
Q_{mmtc}
per-slice Q-heads, separate parameters
accept / reject each pending request,
gNB i reading only its own row
decentralised
execution
7,782 parameters at execution,
the same at every cluster size
Huber loss $\mathcal{L}$, mean over
requests, gradient step
on θ and ϕ_k
per-request target y from
target encoder and heads,
synced every T_{s} episodes
replay buffer $\mathcal{D}$,
minibatch $\mathcal{M}$
store step
centralised
training